

**Matching the Method to the Problem: Machine Learning and Deep
Learning in Real-World Applications**

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Machine learning and deep learning are often discussed as though the newer method is simply the better one. The distinction between them, however, is not a ranking. Machine learning is the broad field in which algorithms learn from data to make predictions without being explicitly programmed, and in traditional machine learning a human selects the features that the algorithm will use. Deep learning is a subset of that field which uses neural networks with many layers to discover those features automatically from raw data (LeCun et al., 2015). The practical question facing a practitioner is therefore not which approach is more advanced, but which one fits the problem in front of them.

This report examines two examples drawn from the course lesson. The first is customer churn prediction using a support vector machine, a traditional machine learning approach. The second is image recognition using a convolutional neural network, a deep learning approach. For each, it identifies a real-world application, explains why the chosen approach suits the problem, and explains why the alternative approach would not.

Customer Churn Prediction With Traditional Machine Learning

Real-World Application

Subscription-based businesses, and telecommunications providers in particular, depend on knowing which customers are likely to cancel their service before they actually do. Acquiring a new subscriber is considerably more expensive than retaining an existing one, so operators run churn models against their customer records each billing cycle and route the highest-risk accounts to a retention team. The model is trained on structured account data that the company already collects: tenure with the company, contract type, monthly charges, payment method, service add-ons, and the number of recent support contacts. A support vector machine learns from historical records in which the outcome is already known, and then scores current customers on their likelihood of leaving. The score is not the end product; it feeds a business decision about which retention offer, if any, to extend.

Why Traditional Machine Learning Is Suitable

The first reason is the nature of the data. Churn data arrives already structured, and the columns are themselves meaningful business quantities. There is no hidden representation waiting to be discovered underneath a value such as "month-to-month contract" or "three support calls in sixty days." The feature engineering that deep learning automates has, in this case, already been done by the business itself in the act of recording the data.

The second reason is interpretability. The purpose of the prediction is to trigger an intervention, and the correct intervention depends entirely on the reason for the risk. A customer flagged because of repeated service complaints needs a technical fix; a customer flagged because a competitor's promotion undercuts their monthly rate needs a pricing response. A model that outputs risk without an accompanying explanation cannot drive that decision. Rudin (2019) argues that in high-stakes settings, interpretable models should be preferred outright rather than treated as a compromise, and retention decisions that affect pricing, service, and fair treatment of customers fall into that category.

The third reason is proportionality. Traditional models of this kind train in minutes on ordinary hardware, can be retrained frequently as market conditions shift, and are straightforward to audit. Empirical comparisons continue to find that tree-based and margin-based methods match or exceed deep neural networks on typical tabular datasets (Grinsztajn et al., 2022), so the simpler approach is not a concession in accuracy.

Why Deep Learning Is Not Suitable

Deep learning's defining advantage is its ability to build a hierarchy of features from raw, unstructured input, learning progressively more abstract representations layer by layer. That advantage depends on structure existing in the raw signal for the network to exploit, such as the spatial arrangement of pixels or the sequential order of words. A row of billing data has no such internal structure. There is no finer-grained pattern beneath a monthly charge for a network to decompose, so the capability that justifies the added complexity simply does not apply.

The costs, meanwhile, still apply in full. Deep learning generally requires far larger volumes of data and considerably more computational power than traditional machine learning (Goodfellow et al., 2016), and a network with many parameters trained on a comparatively narrow table of customer attributes is prone to overfitting. It would also surrender the explanation that made the prediction actionable in the first place, replacing an auditable model with a black box at exactly the point where a company must be able to justify how it treats its customers. The result would be a system that costs more to build and run, is harder to defend, and is unlikely to predict any better.

Image Recognition With Deep Learning

Real-World Application

Diabetic retinopathy is one of the leading causes of preventable blindness among working-age adults, and it is detectable well before symptoms appear by examining photographs of the retina. Detection requires trained specialists, who are scarce relative to the number of patients with diabetes who should be screened annually. Gulshan et al. (2016) addressed this gap by training a deep convolutional neural network on a large collection of retinal fundus photographs graded by ophthalmologists, and reported sensitivity and specificity comparable to that of the human graders. The network is not given measurements of lesions; it is given the image itself and learns what to look for.

Why Deep Learning Is Suitable

The input here is raw pixel data, which is precisely the kind of unstructured signal deep learning was developed to handle. The diagnostic evidence consists of subtle visual features such as microaneurysms, small hemorrhages, and hard exudates, which vary in size, position, and appearance and are photographed under differing conditions of illumination, camera hardware, and pupil dilation. A convolutional network learns a hierarchy from this raw input, detecting simple edges and gradients in early layers and combining them into lesion-like shapes and then into an overall severity assessment in later layers (LeCun et al., 2015). Because convolution

applies the same learned filters across the whole image, the network recognizes a lesion wherever it appears rather than needing to be told where to look.

The conditions that make deep learning expensive are also satisfied in this domain. Large sets of expertly graded images exist, and performance in image recognition has been shown to improve substantially with data and computational scale since convolutional networks first demonstrated their advantage on large-scale image classification (Krizhevsky et al., 2012). Finally, the practical value is a genuine fit for what an automated system does well: applying a consistent standard at a scale and speed that scarce specialists cannot match, in settings where the alternative is often no screening at all.

Why Traditional Machine Learning Is Not Suitable

Traditional machine learning would require a human to specify the features in advance, and for retinal images that means hand-engineering detectors for each type of lesion. This is substantially the approach that computer vision pursued for decades, and its persistent weakness was generalization: a hand-tuned detector that performs well on images from one camera under one lighting condition tends to degrade on images from another. The knowledge required is also difficult to extract, because clinicians recognize these patterns through trained visual judgment rather than through a set of measurements they can fully articulate.

Supplying raw pixels directly to a support vector machine or a logistic regression is not a workable alternative either. Such a model treats its inputs as an unordered list of independent values, so it has no notion that one pixel sits beside another or that a cluster of dark pixels forms a shape. A single one-thousand-by-one-thousand image becomes a million features with none of the spatial relationships that carry the actual meaning, and a lesion appearing in a slightly different position registers as an entirely different input. The information the model needs is present in the image, but the algorithm has no means of organizing it.

Comparing the Two Decisions

The two cases point in opposite directions for consistent reasons, summarized in Table 1. The determining factors are the nature of the data, whether meaningful features already exist, the

volume of data available, and whether the decision requires an explanation.

Table 1

Factors Determining the Choice Between Traditional Machine Learning and Deep Learning

Factor	Churn prediction (SVM)	Retinal screening (CNN)
Nature of the data	Structured account records	Raw pixel images
Feature engineering	Already done; columns are the features	Infeasible by hand; learned by the model
Data and compute needed	Modest; trains on ordinary hardware	Large graded image sets; substantial compute
Explanation required	Yes; the reason determines the intervention	Supporting evidence helpful, but the grade is the product
Result if reversed	Higher cost, lost explanation, no accuracy gain	Brittle hand-built detectors that fail to generalize

Note. Both examples are drawn from the course lesson on machine learning and deep learning.

The course lesson offers a useful summary of the distinction: traditional machine learning is like giving a student a checklist of things to look for in an email to decide whether it is spam, whereas deep learning is like teaching the student to read and letting them work out from experience what makes an email spam. A checklist is efficient and transparent when we already know what belongs on it, which is the situation with churn data. When the defining evidence is something practitioners recognize but cannot fully write down, as with a retinal lesion, the checklist cannot be written and the reader is required.

Conclusion

Neither approach is superior in the abstract. Traditional machine learning is the stronger choice when the data is already structured, the meaningful features are known, the dataset is modest, and the decision must be explainable. Deep learning is the stronger choice when the input is raw and unstructured, the relevant patterns are too subtle or too numerous to specify by hand, and enough labeled data exists to learn them. Applying deep learning to churn prediction would add cost and opacity without adding accuracy, and applying traditional machine learning

to retinal screening would demand feature engineering that has repeatedly proven inadequate. The professional skill is not mastery of the more sophisticated technique; it is diagnosing which of these two situations a problem belongs to before selecting a tool.

Note on the Use of Artificial Intelligence

Claude (Anthropic) was used to assist in organizing the structure of this report and refining its wording. The selection of examples, the reasoning about suitability, and the conclusions are my own, and all cited sources were reviewed for accuracy.

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